Design Activity 2 : Robobee

MAE 3270

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```
% Clear workspace and command window, and close any open figures
clear; clc; close all
```

Define constants and constraints

```
E = 2.4e9;  % Young's modulus of Polyimide film (material "PI (unfilled)" in CES)

% Design requirements
k_hinge = 1.57e-6;  % [Nm/radian] desired hinge stiffness
L_max = 150e-6;     % [m] Maximum hinge length
w_max = 2e-3;       % [m] Maximum hinge width
t_max = 12.7e-6;    % [m] Maximum hinge thicknes (thickness of Kapton film)
```

Compute moment exerted by flapping wing

In this example, we will model the wing as a flat plate. A drag force is created at the center of pressure due to the aerodynamic forces during flapping. We will consider only the drag force (the flow perpendicular to the plate).

```
% Wing parameters
flap_freq = 160;          % [Hz] flapping frequency
wing_area = 40e-6;        % [m^2] wing surface area
y_cop = 6e-3;             % [m] distance from hinge to wing center-of-pressure
theta_max = deg2rad(35);  % [rad] Maximum wing flapping angle
c_D = 1.28;               % Drag coefficient for flat plate

rho = 1.225;  % [kg/m^3] density of air

% Compute max. wing flapping velocity, assuming sinosoidal motion
v_flap_max = 2*pi*flap_freq*y_cop*theta_max;

% Drag force on the wing
F_D = 1/2*rho*c_D*wing_area*v_flap_max^2;

% The force acts at the center of pressure, located x_cop from the hinge
M = y_cop*F_D;  % Bending moment supported by the hinge

fprintf('Moment: %.2e N-m', M)
```

```
Moment: 2.55e-06 N-m
```

Estimate fatigue life of flexural hinge

WRITE YOUR CODE HERE

```
% Define parameters for the fatigue life equation (Dowling, Eqn. 9.7)
% sigma_a = sigma_f * (2*N_f)^b
sigma_f = 1168e6;  % [Pa]
b = -0.159;        % exponent

% Define initial hinge geometry parameters
% current dimensions that give 2 min
t = 7.5e-6;     % hinge thickness
w = 1.25e-3;    % hinge width
L = 70e-6;      % hinge length

c = t/2;        % distance used for bending moment eqn
I = w*t^3/12;   % moment of inertia

% Optimized hinge parameters here
t = 9e-6;       % hinge thickness    // greater than original to increase life
w = 1.65e-3;    % hinge width        // greater than original to increase life
L = 140e-6;      % hinge length      // longer than original to lower stiffness

c = t/2;        % distance used for bending moment eqn
I = w*t^3/12;   % moment of inertia


% Calculate max stress in the hinge based on the moment M
s_max = M*c/I;% MY CODE HERE

% Estimate fatigue life based on sigma_a = s_max
% eqn for life in cycles for mean stress = 0 (not Morrow/SWT stuff)
N_f = 0.5*(s_max/sigma_f)^(1/b);  % MY CODE HERE
```

Verify that your design meets the requirements

```
if abs(E*t^3*w/(12*L)/k_hinge - 1) < 0.1
    disp('Stiffness requirement: ... PASSED');
else
    disp('ERROR: Hinge stiffness differs by more than 10%');
end
```

```
Stiffness requirement: ... PASSED
```

```
if (t > 0 && t < t_max) && (w > 0 && w < w_max) && (L > 0 && L < L_max)
    disp('Geometry requirements: ... PASSED');
else
   disp('ERROR: Geometry out of bounds')
   fprintf('  t = %.2e m, t_max = %.2e m', t, t_max);
   fprintf('  w = %.2e m, w_max = %.2e m', w, w_max);
   fprintf('  l = %.2e m, l_max = %.2e m', L, L_max);
end
```

```
Geometry requirements: ... PASSED
```

```
fprintf('Fatigue life N_f = %.2e (cycles)', N_f)
```

```
Fatigue life N_f = 1.09e+06 (cycles)
```

```
fprintf('Fatigue life T_f = %.2f (minutes)', N_f / flap_freq / 60)
```

```
Fatigue life T_f = 113.75 (minutes)
```

```
if (N_f / flap_freq / 60) > 30
    disp('Fatigue life requirement: ... PASSED')
else
    disp('ERROR: Fatigue life requirement not met.')
end
```

```
Fatigue life requirement: ... PASSED
```

My reflection:

From our equations, I found that

- stiffness was proporttional to width, thickness^3, and 1/length
- maximum stress was proportional to 1/width, 1/thickness^2
- fatigue life was proportional to width, thickness^2, and sigma_f

I used these relations to guide my decisions on how to choose which direction each parameter was changed (greater or smaller) for the optimized design, and it worked efficiently. This was a cool synthesis of the design process + lecture content + real design problems!